What do characteristic words reveal about irony detection?

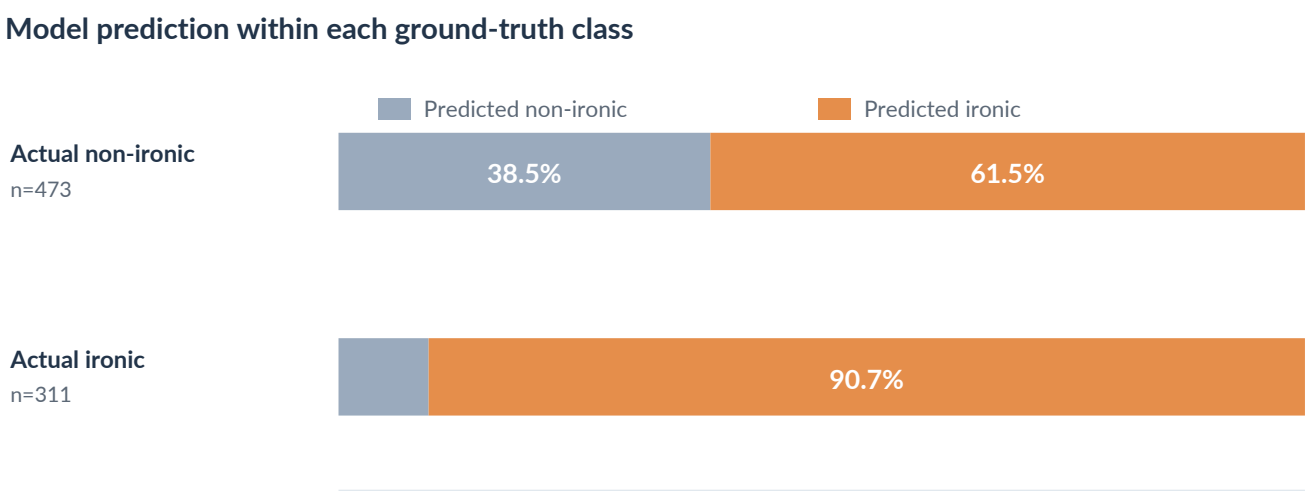
Hypothesis 2 report · Word clusters, prediction gaps, and controlled text edits

Research question. How are particular words associated with Twitter-RoBERTa's irony predictions, and how sensitive are those predictions when the words are changed?

Hypothesis. Characteristic words show different patterns of predicted irony in ironic versus non-ironic tweets; editing those words may alter the model's outputs.

Experiment. We use the pretrained cardiffnlp/twitter-roberta-base-irony classifier on the TweetEval **irony/test** split: **784 tweets** (311 labeled ironic, 473 non-ironic). Word analyses use tweet-level probabilities; no model retraining.

FIGURE 1 · Baseline prediction distribution on the test set



59.2%
Overall accuracy

61.5%
Non-ironic tweets predicted ironic

90.7%
Ironic tweets predicted ironic

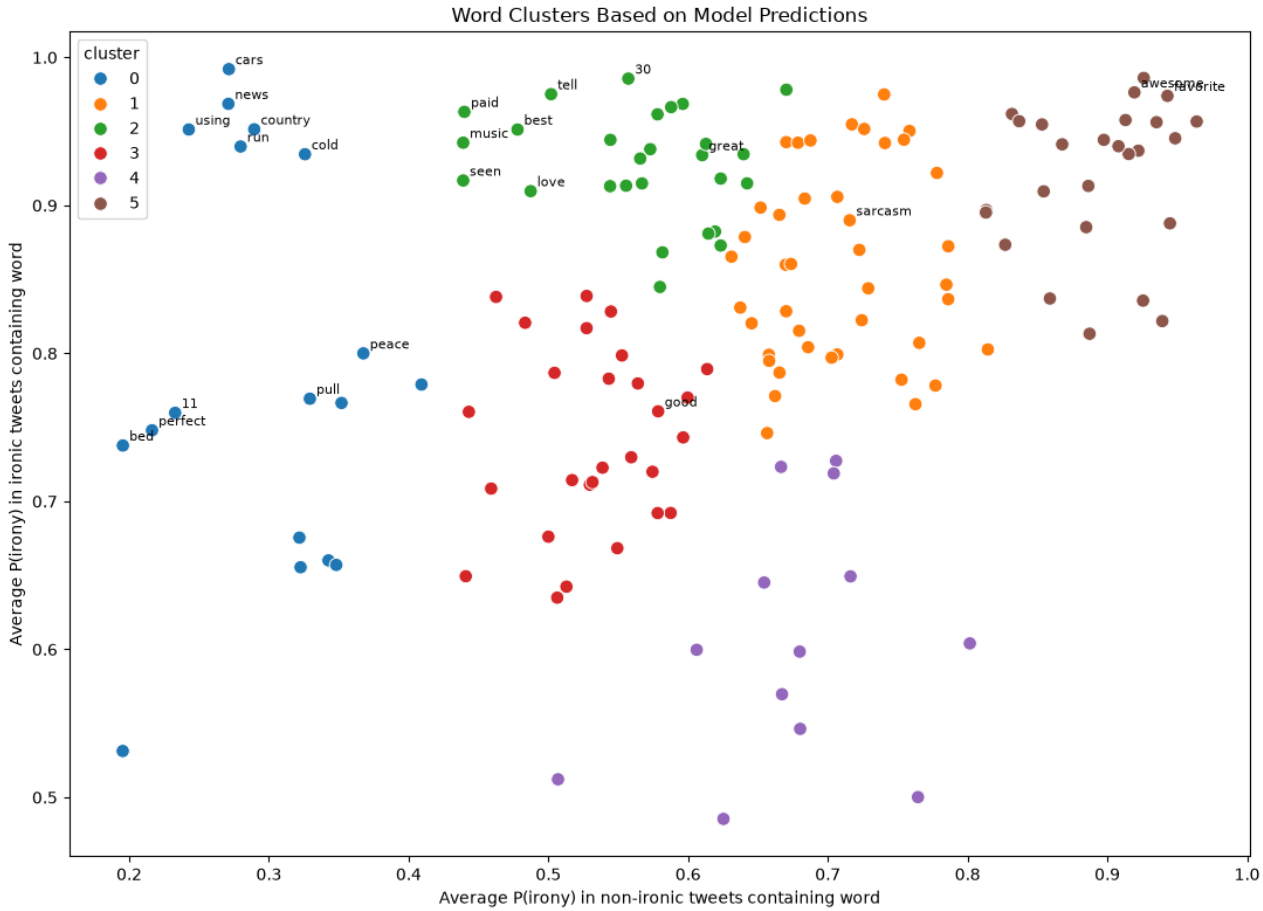
Reading the chart. The model detects 282 of 311 ironic tweets, but also classifies 291 of 473 non-ironic tweets as ironic. This baseline matters when interpreting individual word associations.

Scope. This is an **irony** classifier, not a sentiment classifier. The analyses below investigate black-box behavior: what the model predicts for texts, rather than the word representations it has internally learned.

Word clusters summarize prediction patterns

Each eligible word is described by two features: mean **P(irony)** for **ironic** tweets containing it, and mean **P(irony)** for **non-ironic** tweets containing it. The notebook applies K-means to those two-dimensional summaries.

FIGURE 2 · Six clusters from the word-association features



Axes show **mean model probability for tweets containing each word**, split by true tweet label. Color indicates the notebook's assigned K-means cluster; it is not an internal embedding map.

Eligible vocabulary	Words clustered	Chosen K	Silhouette at K=6
781 words	154 words	6 clusters	0.383

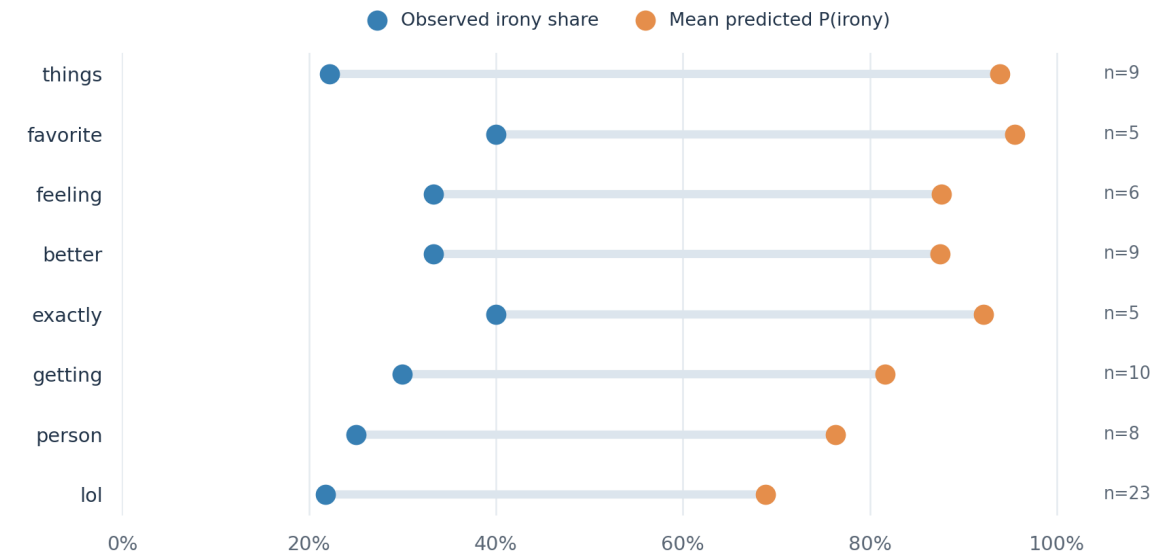
Interpretation. Words such as **great**, **love**, **perfect** and **sarcasm** appear in different regions of the scatter plot. This indicates different average model responses across tweets containing those terms, **not** that the model assigns those probabilities to isolated words.

Limitation. The vocabulary comes from binary word occurrence after English-stopword filtering and requires at least two appearances in *each* true-label group. Six clusters were fixed for the analysis; the reported silhouette is higher at K=7 (0.427), so K=6 is a descriptive choice rather than an established optimum.

Some words accompany strong overprediction

For each word, compare the **observed share of labeled-ironic tweets** containing it with the **average predicted P(irony)** over those tweets. The difference is a **prediction gap**—a descriptive association, not the causal effect of the word.

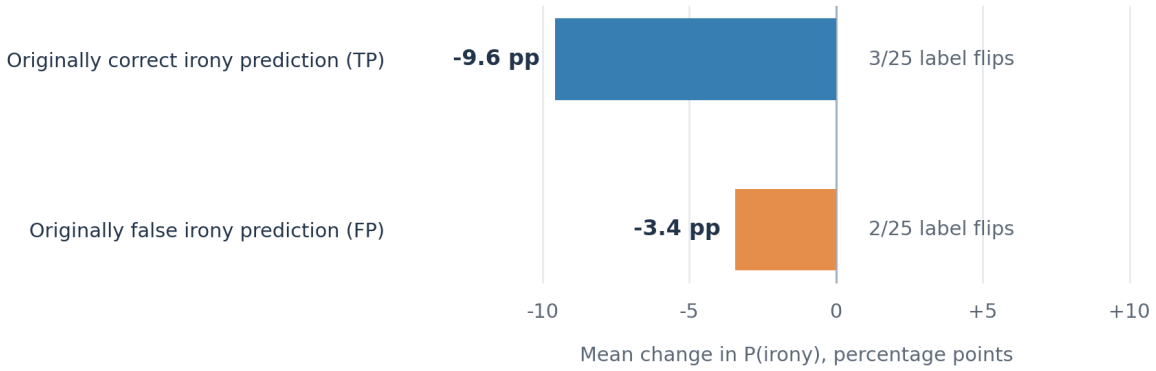
FIGURE 3 · Selected words with the largest positive prediction gaps



Blue: observed ironic share. Orange: average predicted P(irony). The figure shows eight of the notebook's highest positive-gap words. **n** is the number of tweets containing the word.

Examples. For **"things"** (n=9), 22.2% of its tweets are labeled ironic, while mean P(irony) is 93.9% (gap **+71.7 percentage points**). For **"lol"** (n=23), the corresponding values are 21.7% versus 68.9% (gap **+47.1 pp**). Small word-specific samples limit interpretation.

FIGURE 4 · Changing one word in 50 selected TweetEval tweets



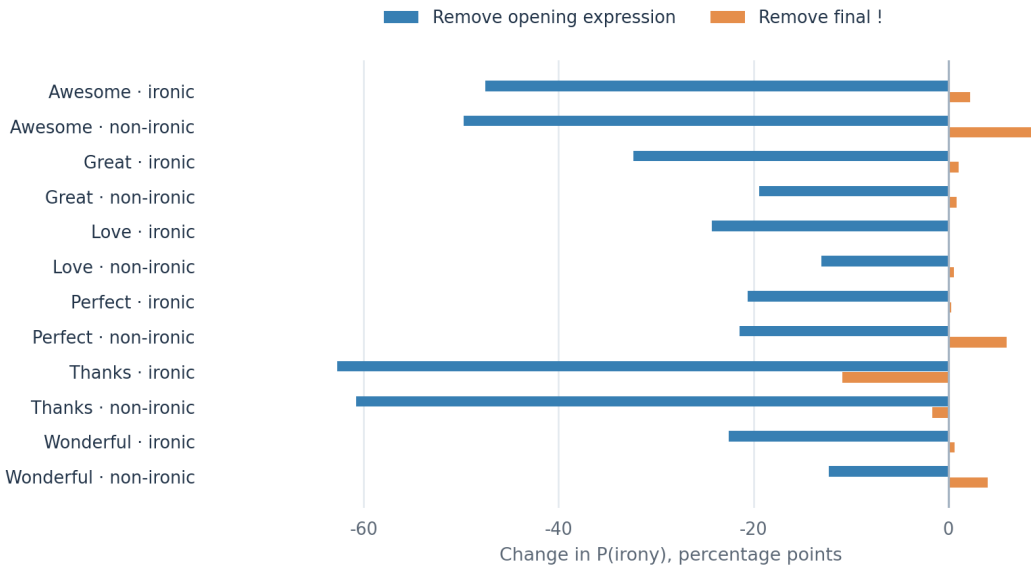
Among **25 originally true-positive** irony predictions, the mean probability change after replacement is **−9.6 pp** and **3** predictions flip class. Among **25 false positives**, it is **−3.4 pp** with **2** flips.

Caution. These examples were selected from tweets already predicted ironic; substitutions include "sarcasm"→"comment", "love"→"like" and "yay"→"well". Some edits change the meaning or spelling of a hashtag. The sample was not random across all test tweets.

The same positive word can trigger irony in either context

Six positive expressions were each tested in an **intended ironic** and **intended non-ironic** sentence. For each sentence, the notebook compared the original, removal of the opening expression, and removal of the final exclamation mark.

FIGURE 5 · Sensitivity in 12 constructed sentence pairs



Across these 12 sentences, removing the opening expression changes P(irony) by **32.2 pp on average in absolute value**; removing the exclamation mark changes it by **3.1 pp**. All 12 original sentences are predicted ironic—even those intended as non-ironic.

Three sentence-level examples

CASE 01 · Intended ironic

Original: Thanks, you lost my luggage!

Without expression: You lost my luggage!

75.7% → 13.0%
−62.7 pp

CASE 02 · Intended non-ironic

Original: Awesome, the server finished the backup before the deadline!

Without expression: The server finished the backup before the deadline!

88.2% → 38.5%
−49.8 pp

CASE 03 · Intended non-ironic

Original: Great, my flight arrived early!

Without expression: My flight arrived early!

97.2% → 77.9%
−19.4 pp

Overall finding. Word associations, real-tweet substitutions and constructed examples all reveal strong sensitivity to wording. They do **not** establish what the model internally learned or isolate the effect of a word from changes in context, punctuation, or meaning.